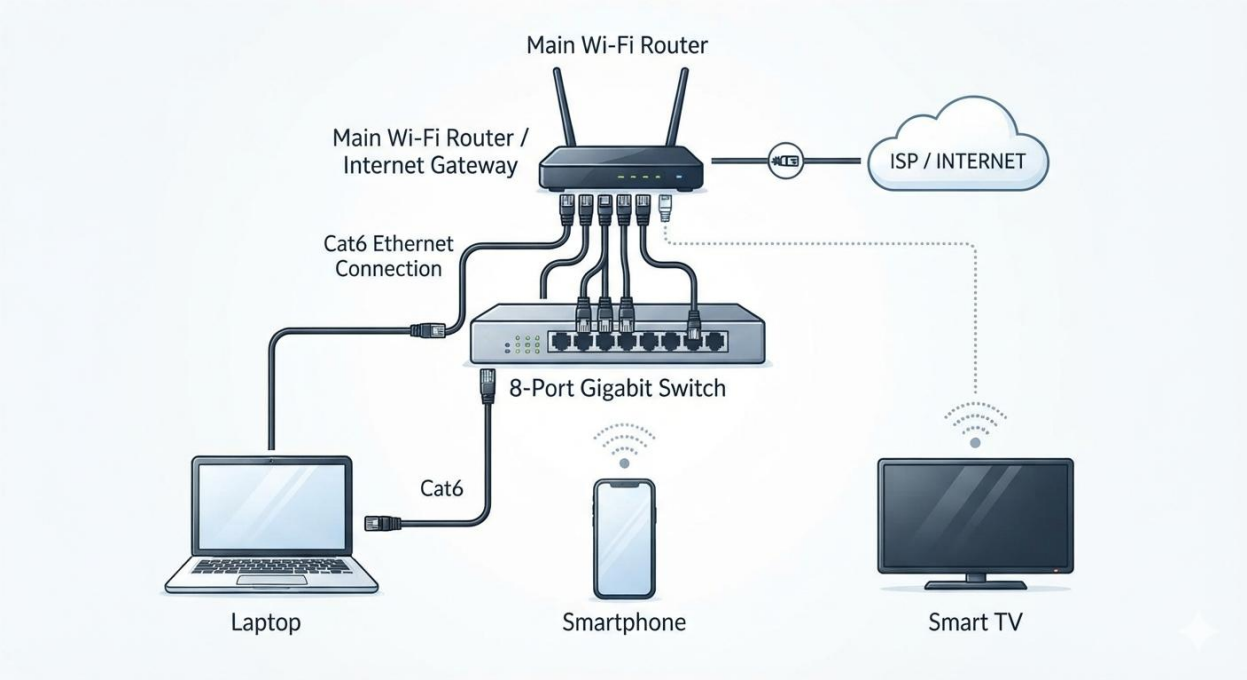
1. **Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.**



- **Laptop**: Connects via a **Cat6 Ethernet** cable to the Gigabit Switch.
- **Smartphone:** Connects **wirelessly (Wi-Fi)** to the Main Wi-Fi Router.
- **Smart TV: C**onnects **wirelessly (Wi-Fi**) to the Main Wi-Fi Router.

2. **Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).**

Network Device	Real-World Use	Daily Example
Router	Connects devices to the internet	Using YouTube over home Wi-Fi
Switch	Connects multiple wired devices in a LAN	LAN gaming between PCs
Hub	Connects multiple devices in a small network	Sharing files between old college PCs
Access Point (AP)	Provides Wi-Fi to wireless devices	Connecting your phone to Wi-Fi for Instagram/WhatsApp

3. **Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).**

- When I order food on Zomato using my phone, the data first travels through the **Access Point**, which connects my phone **to Wi-Fi**. Then it goes to the **Router**, which sends the data from my local network to the internet. Along the way, **Switches** help forward the data between **network devices** and **servers**. Finally, the **Zomato server** receives my order and sends the **response** back through the same network path to my phone.

Phone → Access Point → Switch → Router → Internet → Zomato Server

4. **Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup.**

1. Wi-Fi to Ethernet Adapter

- **How it works:** The adapter connects to the Wi-Fi router wirelessly and provides an Ethernet port. Connect the gaming PC to it using a short Cat6 cable**.**

2. Powerline Adapters

- **How it works**: One adapter connects to the router, and the other connects near the gaming PC. They use the home's electrical wiring to carry the internet connection, with Ethernet cables connecting both devices.